

YUKAI WANG

PhD student | Cho Chun Shik Graduate School of Mobility | KAIST

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RESEARCH PROFILE

Traffic world models, multi-agent trajectory learning, and safety assessment. My research uses drone-recorded trajectories to study interaction dynamics, self-conditioned rollout, and the relationship between predictive accuracy and decision usefulness.

EDUCATION

PhD student | 2023 - present

Korea Advanced Institute of Science and Technology (KAIST)
Cho Chun Shik Graduate School of Mobility | Advisor: Prof. Tiantian Chen

Master's degree | 2020 - 2023

Southeast University
School of Transportation | Advisor: Prof. Zhiyuan Liu

Bachelor's degree | 2016 - 2020

Southwest Jiaotong University
Mao Yisheng Honors College

DOCTORAL RESEARCH | KAIST, 2023-PRESENT

World models from drone trajectories

Research project · Working manuscript

Develop graph-conditioned recurrent state-space models with neural ODE motion decoding; evaluate multimodal prediction and observation-grounded versus self-conditioned rollout on drone trajectory data.

Understanding why closed-loop rollouts fail

Working manuscript

Design paired action/graph source-switch diagnostics and temporal controls to investigate distribution shift and early error compounding in autoregressive multi-agent rollouts.

From predictive validity to decision utility

Working manuscript

Build recording-disjoint, future-free candidate-ranking audits with fixed candidate banks, analytic baselines, leakage checks, rank fidelity, and reference-relative regret.

ProSafeAV: world models meet tail-risk modeling

TRB Annual Meeting 2025 · Ongoing development

Study world-model-based reinforcement learning with traffic-conflict indicators and extreme value theory; develop and evaluate the approach in CARLA-based simulation. Conference work listed at TRB 2025.

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ADDITIONAL ONGOING RESEARCH

Safety assessment from forecasted trajectories

Research in progress

Develop leakage-controlled pipelines from predicted trajectories to traffic-conflict indicators, event-level diagnostics, and tail-risk analysis.

Behavioral validity and cross-site generalization

Research in progress

Investigate layered behavioral validity in calibrated microsimulation and cross-site evaluation of graph-based and autoregressive traffic models.

JOURNAL PUBLICATION

Ziyuan Gu, **Yukai Wang**, Wei Ma, and Zhiyuan Liu (2024). A joint travel mode and departure time choice model in dynamic multimodal transportation networks based on deep reinforcement learning. *Multimodal Transportation*, 3(3), 100137. doi:10.1016/j.multra.2024.100137

CONFERENCE CONTRIBUTIONS

Yukai Wang, Tiantian Chen, and Sikai Chen. ProSafeAV: Enhancing Proactive Safety Performance of AVs using World Model-Based Reinforcement Learning and Extreme Value Theory. *TRB Annual Meeting 2025*. [Record]

Yukai Wang, Tiantian Chen, and Kitae Jang. A Synergistic Approach to Real-Time Crash Risk Estimation at Signalized Intersections: Integrating Learning-Based Anomaly Detection with Extreme Value Theory. *Conference in Emerging Technologies in Transportation Systems (TRC-30)*, September 2-4, 2024. [Record]

INDUSTRY EXPERIENCE

Data Analyst Intern | Shokz | April - July 2023

North America Marketing Department

SELECTED AWARDS

2021 | Top 20, AI Competition. RLChina Reinforcement Learning Summer School.

2019 | National First Prize. 14th National Competition of Transport Science and Technology for Students.

2019 | Honorable Mention. Interdisciplinary Contest in Modeling.

2018 | National First Prize. China Undergraduate Mathematical Contest in Modeling.

2017 | National First Prize. CSEE Cup National University Students Electrical Mathematical Modeling Competition.

METHODS AND TOOLS

Graph neural networks; recurrent state-space models; neural ODEs; autoregressive models; reinforcement learning; surrogate safety measures; extreme value theory; recording-disjoint evaluation and bootstrap diagnostics. Python, PyTorch / PyTorch Lightning, CARLA-based simulation, Git, and LaTeX.

Working manuscripts and ongoing studies are listed as research experience, separately from published articles and conference records. Project figures and current descriptions: yukiwong.github.io/research/.